

School of Physics, Engineering and Technology

ELE00040C

Assessment 2025/26

Summary Details

This assignment (Coursework) contributes **100%** of the assessment for this module, and consists of four components:

- Project specification (Group, 10%)
- Initial group Presentation (Group, 10%)
- Individual Final Report (Individual, 70%)
- Project demonstration and presentation (Group, 10%)

The deadline can be found in the [Statement of Assessment](#). Please try and submit early as any late submissions will be penalised. Assessment information including late penalties is given in [the Statement of Assessment](#).

The **Project specification (Group)** and the **Final Report (Individual)** are written reports. Each report should contain a cover page with the names of contributing authors plus any other details provided in section 5 of this document. Both reports should be a PDF file, use A4 page size with a minimum of 1.5 cm margins all round, and a minimum of 11-point text single line spacing, paragraphs should be justified to both margins. Full details of what to include are provided in section 5. Both reports will be submitted via the **VLE**.

The **Initial group presentation** and the **Project demonstration and presentation** will be in person and arranged by your project supervisor.

Academic Integrity

It is your responsibility to ensure that you understand and comply with the University's policy on academic integrity. If this is your first year of study at the University then you also need to complete the mandatory Academic Integrity Tutorial. Further information is available at [Academic integrity - Studying at York, University of York](#).

In particular, please note:

- Unless the coursework specifies a group submission, you should assume that all submissions are individual and should therefore be your own work.
- All assessment submissions are subject to the Department's policy on plagiarism and, wherever possible, will be checked by the Department using Turnitin software.

Engineering Project 1 - 2025-2026

Medical device development: A prototype wearable medical device to identify tremors

Your engineering project has been designed so that you may apply the technical knowledge you have acquired in other modules to solve a challenging problem. To achieve this you will develop methods in research, project management, resource allocation, effective team planning, scheduling of tasks, documentation control, communication and delivery. This will involve you working within a team to manage the design process and other activities; culminating in the presentation of your final product in a group presentation and an individual project report. The project aims to enhance your skills in creative thinking, analysis and implementation within the context of an engineering application. A full list of learning outcomes and graduate skills can be found in the module catalogue.

1. Assignment Brief and Structure

This project has two main challenges:

Challenge 1 is the specification, design, and construction of a wearable medical device. The device must be able to measure a range of tremors of differing frequencies and characteristics. The tremors being measured will be associated with Parkinson's disease and another commonly misdiagnosed neurological disease. Using these measurements for Rest, Postural, Kinetic and Action tremors, as appropriate, the device should be able to differentiate between Parkinson's and one of the following commonly misdiagnosed conditions; Chorea/Dyskinesias or Essential tremors.

Challenge 2 focuses on testing your device. To do this you will need to simulate a characteristic tremor that can be measured using your device. A mechanical tremor actuator will be designed and manufactured. This process will include different aspects of computing, electronic and mechanical design.

While following the general structure outlined below, each group is encouraged to innovate in all aspects of the project including wearable device design and the mechanism choice for the mechanical actuator.

This assignment (Coursework) contributes 100% of the assessment for this module, and consists of four components, spread throughout the module and weighted as follows:

- Week 5: Project specification (written report) (Group, 10%)
- Week 7: Initial group presentation (Group, 10%)
- Week 13: Final Report (Individual, 70%)
- Week 15: Final group presentation and product demonstration (Group, 10%).

Please note that the submissions are not anonymous and you will be asked to provide your name and the names of your team members on these documents. Also note that for the project specification the same mark will be given to all group members provided they make some contribution to the submission. (If any group member makes no contribution to a group component, that student's mark will be zero for that component.). Weekly team logs will be kept by each team and submitted to your project supervisor, see section 2.3, and used to gauge each team member's engagement.

2. Assignment Scenario, Outline of activities and Budget

The scenario for this project is that you are a team of product development engineers within a small company. The company is looking to move into the wearable medical device market and has chosen the characterisation of tremors as one of several initial projects. Your company aims to sell the device for use within the medical profession but before the process can move forward with a larger investment a prototype device must be produced. Your aim is to produce an initial version of the wearable device and demonstrate that it meets the appropriate technical specifications. As this is a prototype you can assume that the design may be changed (e.g. reduced in size) after this project is finished but before proceeding to market; this could be discussed in the future work section of your final individual report. You should perform an initial literature search and use the information to create a set of technical specifications for the device. You should also use this research to justify the design choices you make and identify the type of measurements you are proposing. The device must be shown to characterise different types of tremor as provided in the specification and so a suitable testing platform and mechanism must also be developed. Finally, you will also need to consider where within the appropriate medical regulatory framework your device sits and develop a plan for how you would take your prototype device to market.

The company is new and has very limited access to finance so you must keep the costs low and your prototype and testing method must cost £100 or less to produce. Your company has already purchased a significant number of components for other projects which can be purchased at a discount (these are supplied in your Programming and Digital Interface Kit)

- Nucleo STM32 microcontrollers, £8
- Accelerometer, LIS30DH - single axis £3, or LSM303D £6 - 3 - axis.
- Micrometal gear motors, either 20:1 or 50:1 gearing ratio £3 (please ask in the EMF for these).

If you wish to employ different components (such as an alternative microcontroller or accelerometer) this must be justified (for example by reference to specific technical specifications) and included in the projected and final budgets for the project. All components and other materials must be purchased and associated costs recorded in the Engineering Ordering System (EngOS) spreadsheet. You will find access instructions on each table in the workshop.

2.1 Your project team

You will be working in your supervision group following the principles covered in the associated project management lectures. The aim of the group work is to give you practical experience of working in teams on a relatively challenging task. A good group will work together consistently throughout the project period, will assign specialist roles or areas of responsibility to the group members according to their strengths, have good procedures and management in place, and will allocate and monitor individuals' workloads and contributions throughout the process.

Even though you are asked to submit an individual report at the end of the project to demonstrate your understanding and progress, you should not think of this assignment as several individual projects – but rather as *one coordinated activity*, spread out between multiple people, using common code, designs and equipment. This will give you good experience for the ways in which companies work.

2.2 Technical Research

This assignment scenario has been set up to emulate the conditions that you might find in a company where a small team of engineers is given a complex technical task to carry out, but they do not currently have expertise in all the required technical areas. In order to carry out this task effectively, your group will need to rely on your existing knowledge and develop new expertise as follows:

- Review your understanding of the STM32 microcontroller, how it is programmed and how to connect and collect data from the accelerometer.
 - You may decide to use an alternative microcontroller in which case you will need to allocate time to learning how that works and is programmed.
- Determine how to record data generated by your microcontroller.

- Review appropriate literature and research to
 - Determine the characteristics of tremors associated with Parkinson's Disease, essential tremor or Dystonia.
 - Identify the needs of the user in order to justify design choices.
 - Determine an appropriate mechanical mechanism to mimic the characteristics of the tremors identified in your research.
- Learn how to analyse the data produced by the accelerometer to characterise the frequency and motion of the wearable device.

In deciding on the specification for your device,

- You should aim to strike a balance between engineering ambition and challenge,
- Ensure that you can get something working for the demonstration,
- Producing an exciting product and demonstrating that it meets the technical specifications.

2.3 Supervision and Document Control

Each team will have an external project mentor (typically your academic supervisor). The role of the mentor is to guide you through the management of your project so that you finish on time. Feel free to discuss the technical specification of the project with your supervisor but where additional technical input is required you will be directed to the workshop technical staff or the module lead. You should schedule meetings with your mentor as appropriate, I suggest scheduling them in weeks 2, 3, 5, 7, 8 and 10. Remember, part of working in a professional environment is developing appropriate meeting etiquette. In the future you will be expected to initiate and hold meetings with clients, managers and other engineers. Use your supervision meetings to work on this professional skill.

It is the responsibility of your team to:

- Arrange the time and location of the meetings. These will typically be in your academic supervisor's office, they may help you arrange a suitable meeting room, or they may meet you in the workshop. In week 1 meet as a group and arrange an initial supervision meeting for week 2. Your supervisor will also have demands on their time, make sure you ask when they are free to meet.
- Be prepared for the meeting.
 - Before your first meeting you should prepare a project charter. This will require you to have performed some preliminary research, decided on a team structure and expand the outline gantt chart.
 - Ahead of each meeting review your log and assess your progress against the Gantt chart. Be prepared to discuss items highlighted in the log and your progress.
 - Allocate some time in this meeting to arrange the location and time of the following meetings.
- Complete the team log each week, be prepared in your meeting to discuss the ongoing work, problems encountered and your progress against the timeline in your Gantt chart.

Where needed you will work with your supervisor to update plans as the project evolves in order to deliver a tested product in your final group presentation.

Please note you must provide evidence of your contribution to the project in the team log, this will be referenced by you in your project specification and final report.

Document control: Team log and outline Gantt chart templates are provided in your team's google drive folder. Please use these folders to store all of your documents, this will allow your supervisor to track your progress and provide advice. In these logs you should summarise the work that your group has done during that week, individually report on your progress and set targets for the following week. This report should include minutes of any group meetings (and it's strongly recommended that you schedule at least one group meeting per week, these are in addition to your meetings with your supervisor). You will receive feedback on these documents from your allocated project mentor (which is usually your academic supervisor). These documents will also be used as source material for your final individual reports.

2.4 The Impact of Engineering on Society

In the rapidly evolving field of engineering, it is important to address not only the technical aspects but also the societal implications of your innovations. This project has been designed to integrate the five key societal engineering themes that have been defined by the UK Engineering Council:

1. Sustainability,
2. Ethics,
3. Equality/Inclusion,
4. Risk, and
5. Security

These themes help us all to develop an understanding of how our engineering solutions affect the world around us. By considering Sustainability, we ensure that our projects are environmentally responsible. Ethics helps us navigate the moral implications of our technological choices. Equality and inclusion ensure our designs are accessible to everyone, irrespective of their abilities or backgrounds. Predicting Risk prepares us to anticipate potential failures or bad outcomes, and prepare solutions. Lastly, focusing on Security ensures the integrity and safety of our technological solutions.

Assessing these aspects in your projects broadens your learning experience and prepares you to become thoughtful, responsible engineers who can contribute positively to society. We encourage you to integrate these themes into your projects, and they will form a key part of your assessment criteria.

We ask you to consider these issues as a Group, and to report on them in your Initial Group Report (see Section 5.1 below), you may address these by either including them in the relevant sections of the report or by downloading, filling in, and including the table in appendix A. In addition we ask you to reflect Individually on each of these issues in the final report.

3. Checklist of Individual Tasks

The tasks required to successfully complete the project are:

1. Ideation and planning:

- Define the characteristics of the tremors you will be measuring. Remember, you need to differentiate between at least two types of tremors.
- Determine a method for measuring tremors that can be housed in a wearable device, taking into account the requirements of the user.
- Consider how and what data will be recorded.
- Determine a mechanical mechanism that can mimic the characteristic of the tremors.
- Consider opportunities for innovation and specialisation.
- Equality is an important element of the design process. In designing your device are you considering all stakeholders and their needs?
- Consider the security of the data collected by your device.
- Produce a project plan that breaks down the work to be completed, construct an appropriate timeline of activities, plan identifies and mitigate risks.
- What impact does your idea have on society?
 - Determine whether there are any *ethical considerations* that need addressing in your project. Complete the ethics form on the vle; this will aid you in identifying any areas of ethics to address.
 - Does your project plan *equitably distribute* the work between members of the team, and *addresses and manages risks to the project*.

2. Product design:

- Design the electronic components for the wearable device. Don't forget to include a suitable method for
 - collecting tremor data.
 - recording tremor data.
- Design a suitable housing and method for mounting the wearable device.
- Design and implement the logic for the microcontroller.
- Design a mechanical actuator that can be used to mimic tremor characteristics. A prototype stand is provided that can be used to house your actuator, feel free to use this or design your own if you have the time available.
- What impact does your design have on society?
 - Identify any constraints for your design, these can include *environmental* and *sustainability* limitations; *ethical*, *security* and *risk* issues; intellectual property.

3. Product Manufacturing:

- Determine the most appropriate method for manufacturing your individual components.
 - Try to strike a balance between cost and desired finish, e.g. 3D printing is available but can be slow and expensive. Laser cutting is quick and much cheaper.
- Determine appropriate materials for each component.
- Produce CAD files for all the physical components that can be used by the 3D printer or the laser cutter.
- Consider the use of CAD to test the mechanical design prior to the construction of your tremor generator.
- You will be introduced to the manufacturing methods available in the first workshop session in week 1. However, if you are unsure which processes to employ or what is available then please talk with the workshop technical staff or your module coordinator.
- What impact do your manufacturing choices have on society?

- Consider the sustainability of your material choices.

4. Testing and refinement:

- Design and perform tests to demonstrate that the wearable device meets the specification.
- Learn the appropriate statistical methods for analysing the data (a lecture will be provided on this subject prior to this state of the project starting).
- Collect and analyse test data for the wearable device.
- Refine the design of the tremor generator or wearable device based on the results of the tests.

5. Documentation and presentation:

- In your team log, document your project's management plan, design process, technical specifications, and any innovations or specialisations.
- Combine all of the work performed whilst planning into your initial group report (specification) and the initial group presentation.
- Prepare a final presentation demonstrating your wearable device and show that it meets the specifications.
- Write your comprehensive individual report.

4. Time and Project Management

Effective time and project management skills are crucial for the successful completion of any project. This section outlines a structured approach to managing the group project, incorporating important project management techniques and tools.

By engaging with these methods, your group will maximise the chances of a successful outcome, and also gain valuable experience in teamwork, collaboration, and project management.

1. **Project Charter:** At the beginning of your project, create a project charter that outlines the project's aims, scope, deliverables, team roles and responsibilities, and the objectives (major milestones during development). This document will serve as a reference and guide for the team throughout the project. You will need to include some of this in the initial report, along with items 2, 3, and 5 below.
2. **Work Breakdown Structure (WBS):** Divide the project into smaller, manageable tasks and organise them in a hierarchical structure. This helps the team visualise the project's scope, assign tasks to team members, and track progress.
3. **Gantt Chart:** Develop a Gantt chart to schedule the project timeline, outlining the start and end dates for each task. This tool will help your group to monitor progress, identify potential bottlenecks, and ensure that the project stays on track. An outline gantt chart is provided in your groups google drive but this will need updating with tasks specific to your project. Are your tasks equitably distributed between group members.
4. **Regular Progress Meetings:** Schedule regular team meetings (at least one per week) to discuss progress, address any issues or concerns, and facilitate communication among team members. These are in addition to your supervision meetings. During these meetings, use tools such as task boards or Kanban boards to visualise the progress of individual tasks and the overall project. At each meeting, identify actions for each team member for the next week. Take minutes of the meetings to act as a record of your decisions. You should put an entry into your team log for each meeting.
5. **Risk Management:** Identify potential risks and challenges that may arise during the project and develop strategies to mitigate or address them. Monitor and review these risks throughout the project to ensure they are effectively managed.
6. **Peer Review and Feedback:** Encourage group members to review each other's work and provide constructive feedback. This process helps identify areas for improvement, encourages collaboration, and enhances the overall quality of the project. Use design and coding review meetings to check and identify issues with designs before they are implemented.

7. **Time Management Techniques:** Encourage each other to use time management techniques, to optimise their personal productivity and maintain focus on their tasks.

5. Assignment Deliverables

This section summarises the requirements of the assignment, then gives more details on the four individual submissions in the order of their deadlines.

There are four assessed deliverables for the project:

1. A **Project specification (written group report)** which describes your team and its plans.
2. An **initial group presentation (group)**, you will present your project management plan, research, device specification, and any initial designs.
3. A comprehensive **individual project report** detailing the design process, technical specifications, innovations, and specialisations.
4. A **product demonstration and group presentation** showcasing your group's work, including a demonstration of your wearable device, the method used for testing, and appropriate results demonstrating whether the device meets the technical specification you provided.

These deliverables are described in more detail below.

5.1 Initial Report: Project specification (Group submission, 10%)

Write a group report that describes your team and the initial planning for the project. There is a limit of 20 pages for this document and you should not exceed this limit. This length has been chosen to encourage you to use as many diagrams as you like (such as details of your plans, company structure charts, ideas sketches, Gantt chart, tables etc). As a guide to structuring your 20 pages you should aim for approximately 2000 words of written content and use the remaining space for diagrams and figures.

You will be given guidelines and a [marking rubric can be found in Appendix B of this document](#). The report should cover the following items.:

Your Team: Your report will start with the following information and this will be graded as part of the overall presentation. Project name and any logo you designed, a list of the contributing authors (members of the team that worked on the report), expected start and completion dates.

1. **Defining your idea:** The product you develop must provide a solution to an identifiable problem. To achieve this you will need to set clear aims and objectives; these will focus your work on solving the identified problem. Consider who will use your device, both in the medical profession and the patients. Perform a literature search of appropriate research papers and market competitors. Using your research define the problem you are attempting to solve, identify the needs of the end user of the product, and consider the merits and limitations of existing solutions. Work as a team to identify your solution, produce initial sketches and CAD drawings; consider appropriate components and materials for construction. These do not need to be finalised at this stage but the technical requirements should be identified, see technical specification below.

You should include any research in your project specification as a literature review. Your literature review should aim to address the following:

- A summary of existing devices, methods for testing and what problems they are trying to solve. Try to identify general trends in their characteristics that make these good or bad solutions.
- Identify the characteristics of the tremors you wish to measure.
- The research presented in your literature review should help to justify your proposed solution.
- Any research source that is not produced by you in this project should be referenced using a recognised style. See the [university referencing style guide](#).

Things to consider, who will use the device and for what (in a clinical environment, at home monitoring, personal use)? What kind of tremors will the device characterise? How do you know what tremor characteristics to focus on? How innovative is this solution? What other similar types of device exist? How will you test if the device meets your technical specification you have set out?

2. **Project scope:** Describe your initial ideas about your device. Include any initial sketches of the product. Also describe how you intend to test the device. Clearly define what will be delivered at the end of the project. You may also consider what falls outside the scope of the project (e.g. If you are not going to look at some aspects of tremors highlighted by your research). You should justify your design choices and you can refer back to your literature review or introduce new references as required.
3. **Technical Specification including cost:** Describe what your product will do, and how it will operate technically. Remember you are aiming to address an identified problem (e.g. does your device measure tremors in the appropriate frequency range?). This description will help you choose materials and components, if you have had any initial ideas then include these in the technical specification. Your specifications should be justified by appropriate research and you can refer back to your literature review or introduce new references as required.

Describe how your design meets the engineering themes (see section 2.4) of Sustainability, Ethics, Inclusivity, Risk and Security. You need not address all of these in the technical specification and as a minimum for the ethics you should complete the ethics form and have this signed by your supervisor.

You should also detail the target cost for manufacture of your product: provide a breakdown of the approximate costs of any materials or components used. You will still have design and manufacturing choices to make at this point so this will be an approximate budget, you can review and reflect on how well your approximation worked in your final report.

4. **Project Management:** Describe how your project will be organised and managed. You should consider providing a table of responsibilities allocated to individual group members, a work breakdown structure, a Gantt chart with overall timing and milestones, and a risk analysis (this is a good opportunity to address one of the engineering in society key themes). Explain how documents and data will be managed. Give consideration to the ethical implications of the product design process.
 - o **York strengths grid:** For each team member produce a york strengths grid. Identify the location within the grid for each of the 9 york strengths. [Further details of york strengths are given on the website](#). You may include these in an appendix but reflect on your identified strengths and weaknesses when organising the team.
5. **Impact on society:** Have you addressed the impact on society or do you need to complete the table in [appendix A](#)? It is best to address these ideas throughout your technical specification and project planning, but you may also wish to bring them together in one place, clearly identifying where you have looked at each of the 5 key themes.
6. **Overall presentation:** Project title that clearly articulates the content of the report. A list of contributing authors. The report should be the correct length, be free from grammatical and spelling errors, figures should be clearly captioned and referenced in this text, the figures should be of sufficient quality and size that they are legible to the reader. A clear and easy to follow structure that incorporates all components of the project specification. A complete list of references that identifies the work of others used in the project must be provided using a recognised style.

Contribution: This is a team project and in order to be successful it is important that each member contributes equally to the project. Your supervisor will monitor the work of each team member and track their contribution. This will be based on the information in the report, for example named activities on the Gantt chart that have been completed by the time the report is submitted, work in preparing and planning that was presented in supervision meetings, and details of completed work within the team log. If you make no contribution to the report then you will receive a mark of zero for the report.

Roles that could be assigned to various team members include the following:

1. Ideas brainstorming.
2. Literature search, market research and report writing.
3. Time-planning and coordination.
4. Costing the final design.
5. Describing your approach.
6. Producing technical plans such as CAD files.
7. Producing diagrams.
8. Re-drafting and compression to meet word-count limit.
9. Proofreading, checking and editing.

Remember that in a Group Report everyone should be contributing equally, but just one person from each group (selected by the team) should submit the Group Initial Report (in PDF format) to the submission point on the VLE.

Format: Produce a PDF file for your report, using A4 page size with a minimum of 1.5 cm margins all round, and a minimum of 11-point text single line spacing, paragraphs should be justified to both margins.

5.2 Initial group presentation (Group submission, 10%)

The initial project presentation is a group presentation with maximum 12-slides with 2-slides per individual, to be presented in 15 minutes with the following 10 minutes questions/answers.

Below are guidelines on how to approach this task and the elements that should be included to make the best impression. (These elements also form the Marking Scheme for this report; the weight of each component is given.)

1. Format and length (20%):

- The presentation should take a maximum time of 20 minutes and you should aim to use as much of this time as possible.
- You are allowed up to 12 slides and should allow approximately 2 slides per individual.
- Well presented slides that address the presentation content outlined.

2. Presentation content (40%):

- **Introduction:** Introduce your team members, mentioning their roles, qualifications, and expertise.
- **Problem:** Clear outline the problem you are trying to solve indicating the research you performed to arrive at this outline.
- **Concept:** Clearly present your proposed solution and its specification.
- **Plan:** Outline the project plan, providing a timeline for each phase of the project, identify any significant milestones, identify any risks and your plan to mitigate them, describe how you will divide up the tasks within your team.
- **Conclusion:** Summarise the main points of your proposal and explain their benefits.

3. Presentation (20%):

- Each team member present should contribute to presenting some part of the presentation.
- Ensure that the presentation runs smoothly and to time (arrange to practise ahead of time).
- Clearly articulate each of the points listed in the presentation content section.
- Ensure that the slides are clear, the content relates to what is presented and can be easily read by the audience.

4. Question and answer (20%):

- There will be approximately 10 minutes of questions relating to your proposed plan and solution at the end of the presentation.

Keep in mind that this presentation is an opportunity to showcase your group's creativity and professionalism. Be sure to rehearse and refine the presentation to ensure it is polished, engaging, and communicates your proposal effectively.

5.3 Final Project Report (Individual submission, 70%)

The final project report is an individual report. The report should be a maximum of 12-pages long. Although you worked as a group your report is written independently. If you wish to write about the specific work of other group members you may do so and would reference this in the usual way.

5.3.1 Report Length & Purpose

Your report should be no more than 12 pages and should include plentiful diagrams, such as concept sketches, cad designs, pictures of the devices throughout the project. Even though you will have been working on a good deal of technical material it is always important as engineers to be able to summarise a whole project in an executive summary; this report assignment gives you practice and feedback on that skill. You may include appendices. Make appropriate use of appendices, core information should go in the main body of the report and supporting material may be placed in an appendix. For example, you may include your full analysis code for the tremor data in an appendix and put a flowchart describing the process in the main body of the report. More details of the required contents are listed in the marking scheme below.

Note that the page limit relates to the main body of the report so excludes the title page, list of references, and any appendices.

5.3.2 Report Format & Organisation

Produce a PDF file for your report, using A4 page size with a minimum of 1.5 cm margins all round, and a minimum of 11-point text single line spacing, paragraphs should be justified to both margins.

The report should contain a front page with the School, Project Title, Date, Group Number and Supervisor, your name (clearly identified as the report author) and the names of the other group members.

5.3.3 Report Content & Marking Scheme

The expected report contents and presentation quality (together with their weight towards the report mark) are:

- a) **Introduction and background (15%).** Explain the main aims and objectives of the project, outline the problem that the product addresses (reference your project specification if required but avoid cutting and pasting in large blocks of text), provide a breakdown of the work performed. You should include a sufficient amount of information so that a well informed reader can understand the aims of the project, what was delivered, and why it was needed, without reference to other documents.
- b) **Description of wearable device and testing method (10%).** Although this is a group project and you did not work on all aspects of the project you should provide a description of everything your team produced.
- c) **Analysis of Effectiveness & Future Plans (15%).** Describe the achievements of the prototype compared to the specification in the initial report. Explain whether you met the specifications (both technical and projected cost). You should explain how you demonstrated your device meets the technical specification and include the results of any tests with suitable conclusions. Explain any further work that would be needed as a group to complete the product prototype. Consider where in the medical regulator framework this device sits and what work needs to be performed in order to take this product to market. Note that if you

are using the work of other team members then you can include a suitable citation to indicate who performed the work.

d) Individual Contribution (35%)

- a. **Clear statement of individual role and a list of your contributions** to the project. This should include references to your progress statements in the team logs.

Please also include a list of the main roles taken by the other team members.

- b. **Technical detail of individual contributions**

Here, describe all the work that you have personally contributed (or jointly worked on). For example; if you worked on CAD designs of the wearable device enclosure then include details of design and its evolution. If you have managed to produce code and configure hardware, then include details of code. If you constructed the tremor actuator then perhaps include photos. Where testing was performed explain what you did including the key results and conclusions (code, further images and raw data can be placed in an Appendix).

Note that if you are reporting work which you have jointly done with other team members, you must clearly specify your individual contributions to these tasks.

If you have not managed to make as much progress as initially expected, then it is vital that you describe the *progress that you have made, and your plans for future work* in enough technical detail that someone else could complete the work in a well-equipped laboratory. For this, you should contemplate including some of the following as appropriate:

- Flowcharts outlining the logic of a code.
- Written (but as yet untested) code.
- Plans for testing and what a good result would look like.
- Results from simulated data and how you expect this to differ from real data.
- CAD designs and simulations of working models.

e) Conclusions, Reflective Summary and Strengths Analysis (15%) – see sections 5.3.4 and 5.3.5 below.

- You should conclude your report by briefly summarising the key points.
- Also see sections 5.3.4 and 5.3.5 for a description of the reflective summary and strengths analysis.

f) Overall Appearance and Structure (10%)

Your report should have a clear structure with section headings and subsections as appropriate. All work that is not your own should be referenced using an appropriate referencing style. It should be free from grammatical and spelling errors. As a guide you should try to produce something that you would be comfortable passing on to your boss. Follow the recommendations for report writing (available on the wiki) and use plenty of diagrams. Each diagram you include should contain a brief caption and be numbered. Use the number to refer to the diagrams from the body of the report. Where appendices are included these should contain appropriate content. If you make heavy use of acronyms then provide a table of acronyms at the start. You are strongly advised to get some feedback on your report's readability and format before submission.

5.3.4 Details on Reflective Summary

In your Individual Report, you should reflect on and summarise how well your group was structured, and how well it worked in practice.

Reflect on your time management and project planning, especially in the light of any unforeseen circumstances that have occurred since you started the project. How has your group coped with the change, shown resilience, and reconfigured its working operations?

You should draw on your personal progress each week (as documented in the weekly progress reports) as source material for your reflection and include your contribution to the weekly reports as an appendix to your report.

Also include (as an appendix, if necessary) any Gantt chart or Work Breakdown Structure that was produced at the start of the project and comment on your progress against it. Also, comment on how well your Risk Management helped you mitigate any problems that did occur and anticipate and avoid any problems that might otherwise have occurred.

5.3.5 Strengths Analysis

In preparation for future job interviews, it is essential to reflect on your skills development throughout your projects.

One common interview question is, *'Tell us about a recent project that you have been involved in, and how it has developed your skills.'* In fact if you search for common interview questions many can be answered by reflecting on the work you performed in this project.

To help you prepare for this, you are **required to fill your York Strengths grid at both the start and end of this project, and then comment on the progress** as if in a written answer to that interview question. You must include your york strengths initial grids in the Project specification and the updated grid in the Individual report. [Further details of york strengths are given on the website.](#)

This exercise helps you to identify and articulate the skills you have developed, and enables you to clearly demonstrate your growth to potential employers.

5.4 Project demonstration and group presentation (Group submission, 10%)

As a group you should prepare a final presentation that provides details of your device, how it was tested, and a demonstration of its capabilities.

1. Presentation length and format (20%):

- The presentation should be not longer than 15 minutes, use as much of this time as possible.
- Provide a demonstration of your working derived and tremor sensor, either live or as a video. Aim for this to take no more than 5 minutes of the total presentation time.
- Use 8 slides covering the content provided below, approximately 1 to 2 per team member.
- All team members should contribute to the presentation.

2. Presentation content (40%):

- **Introduction:** Introduce your team and provide a brief overview of the wearable device and the tremor actuator. Mention the main functions and any unique features or specifications of your product.
- **Product demonstration:** either live or as a recording demonstrate your wearable device and tremor actuator. Show how it meets the specifications for the product.
- **Achievements and challenges:** Highlight any significant accomplishments or milestones achieved during the project. Discuss any challenges faced and the solutions or strategies employed to overcome them.
- **Conclusion:** summarise the key points of your proposed solution. Highlight how it meets the criteria set out in the specification.

3. Presentation (20%):

- Ensure that the presentation runs smoothly and you clearly articulate each of the points listed in the presentation content section.
- Ensure that the slides are clear, the content relates to what is presented and can be easily read by the audience.

4. Question and answer (20%):

- There will be approximately 10 minutes of questions relating to your proposed plan and solution at the end of the presentation.

6. Submission

Your Reports (in PDF format) should be uploaded to the VLE by the date shown in the statement of assessment.

- For the Project specification (Group) just one person from each group (selected by the team) should submit the report (in PDF format) to the submission point on the VLE.
- For the Final Report (Individual) each of you should submit your own report (in PDF format) to the submission point.
- For the presentations, both the initial and project demonstration, you should arrange with your project mentor a suitable time to hold the presentation. These should take place by the end of the week identified for the presentation.

Any queries can go to the module coordinator (paul.davies@york.ac.uk) or your project supervisor.

Appendix A: Impact of engineering of society assessment

Engineering Council's Societal Category	Ask yourself these questions about your project and write your answers in the next column (also refer to the hints below this table)	Your assessment (20 - 200 words) [if a particular aspect doesn't apply to your project, then declare it so with a reason]
Sustainability 	What environmental impact does your design choice have? <i>E.g., Think about materials used, are you reusing components from other projects, e.g. recycling components. You may wish to refer to UN's sustainability goals, and to specific directives such as ROHS and WEEE.</i>	Your text here
Ethics 	Discuss the ethical implications of your design. <i>E.g., Have you used the ethics checklist to identify ethical implications of your work. Are you using copyrighted material, sourcing materials from trusted suppliers. Are you testing the efficacy of your proposed solution?</i>	Your text here
Equality/Inclusion 	How accessible is your product? Is your plan equitable? <i>Have you considered all stakeholders in your planning and design? This might be broader than just considering the doctor that will purchase the device. How will patients interact with your device? Is it to be used at home? Is your project management plan inclusive and equitable?</i>	Your text here
Risk 	What potential risks have you identified and how does your design address it? <i>Within your group, what could go wrong? How can you mitigate for these risks by careful planning and choices? Did you include a project risk plan in your initial report?</i>	Your text here
Security	What security measures need to be considered?	Your text here



Is this important for your proposed solution? The device may store data on the evolution of a medical condition. How would you ensure this is secure? I.e. could this be accessed easily by third parties. This will be particularly important when considering how data is transferred from the device.

Further Reference

See pages 5-7 of the Quality Assurance Agency for Higher Education (QAA) Subject Benchmark Statement for Engineering, which defines these standards in more detail (N.B. Risk here is called Safety).

[QAA subject benchmark for engineering \(opens a PDF\)](#)

Appendix B: Initial Report/Project Specification Marking Rubric

	<80%	79 - 70	69 - 60	59 - 50	49 - 40	>39
	Excellent	Very Good	Good	Satisfactory	Passing	Failing
Introduction, aims and objectives (5%)	A compelling clinical/engineering context, justifying the need for the project. Clear overarching aims. Objectives are clear, measurable and well mapped to the project. The aims and objectives are ambitious but achievable within the project's timeframe and resources.	A strong clinical/engineering context, justifying the need for the project. Aims and objectives are comprehensive, realistic and well mapped to the project.	Solid clinical/engineering context that justifies the need for the project. Aims and objectives are set. These are reasonable but may lack precision or full consideration of constraints.	Basic introduction with limited context. Aims/objectives are vague or incomplete, with little consideration of project requirements.	Minimal introduction with little clarity. Aims are weak, unrealistic, or disconnected from the project.	Very limited introduction. Aims are unclear, irrelevant, or show little understanding of the project.
Research and literature review (20%)	Critical Synthesis. Expertly compares existing clinical solutions and sensors. Defines appropriate tremor characteristics with high rigour. Extensive range of high-quality peer-reviewed and wider sources used.	Analytical. Demonstrates substantial background reading. Strong comparison of market solutions. Tremor characteristics are relevant and well defined analytically. Justifies project needs through peer-reviewed evidence.	Descriptive/Analytical. Solid range of sources. Some critical comparison of existing tremor devices. Tremor characteristics are appropriate but the description lacks depth.	Descriptive. Sufficient sources but relies on "listing" rather than comparing. Basic tremor data.	Too few relevant sources used, mostly limited to a single technical area. Insufficient wider reading and possible over-reliance on low-quality sources.	Very few references are used and many are irrelevant. Little evidence of meaningful background reading.

Technical specification (30%)	A clear and detailed technical specification of both the device and testing mechanism is presented and fully linked to the research presented. The choice of materials and components is well justified with sound reasoning, alternatives have been considered. Excellent distinction between patient and clinician needs; these are all addressed by the design.	Complete specification of both the device and testing mechanism is presented and well linked to the research. Most of the choices for materials and components are justified with sound reasoning. Clear distinction between patient and clinician needs; there is an attempt to address these in the design.	Technical specification is mostly clear, includes information on both the device and testing mechanism, some justification linked to the research. Some consideration of the materials and components. Users are identified but no clear distinction is drawn between the doctor and the patient's requirements (or focus too heavily on one user).	Technical specification present but generic with some justification but weakly or intermittently linked to the research. Listed component choices with limited justification. A user is identified, typically the patient, no consideration given to other users.	Technical specification incomplete or vague and minimally linked to the research. Choices of materials/component s listed with no justification. Limited consideration of the end user.	Limited or unrealistic specification. No meaningful justification; little or no link to background reading
Project Planning (20%)	Comprehensive and realistic plan with detailed Gantt/WBS. Tasks, milestones, and dependencies are clear, feasible, and well justified in terms of the project aims, objectives and technical specification. Clear team structure laying out roles and responsibilities for each group member. Team roles fully justified through suitable reflective exercise, going beyond York Strengths. Skills/knowledge gaps identified and	Strong plan with clear Gantt/WBS chart. Tasks and milestones mostly realistic, though some detail or justification could be stronger. Clear team structure laying out roles and responsibilities for each group member. Team roles mostly justified through suitable reflective exercise and acknowledges some	Adequate plan with a workable Gantt chart. Covers main tasks, limited detail on timing, or justification. Roles of individual team members identified but lack detail. Some reflection of strengths and justification for assigning team roles (e.g. YS grid, or other suitable reflective exercises).	Basic plan with a simple Gantt chart. Some tasks missing or unrealistic; little evidence of critical thinking about scheduling. Limited evidence that core skills of the team have been considered (e.g. YS present but incomplete).	Incomplete or vague plan. Gantt chart poorly constructed, with major gaps or unrealistic timings. Little to no evidence that core skills of the team have been considered (e.g. YS missing).	Limited or confused plan. Gantt chart unclear or irrelevant, little evidence of structured planning.

	mitigated.	skills gaps with some efforts to address these identified.				
Engineering in Society (Ethics, Risk, Sustainability, Equality/Inclusion, Security) and Costs (15%)	Thorough consideration is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). Identified issues are all addressed appropriately through the design choices, planning, or through well reasoned mitigation strategies. Thorough identification of financial considerations with realistic and fully justified costs.	Clear consideration is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). Most identified issues are addressed appropriately through the design choices, planning, or through reasonable mitigation strategies. Financial considerations with realistic and mostly justified costs.	Adequate consideration is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). Some identified issues are addressed appropriately through the design choices, planning, or through reasonable mitigation strategies. These are limited in depth. Costs listed with few minor omissions and with some justification.	Some superficial or generic consideration is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). The breadth and depth of this consideration is limited. Costs listed with no significant omissions but some figures are unrealistic.	Incomplete or vague discussion is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). Very limited costs with unrealistic figures.	Missing or irrelevant discussion is given to the engineering in society categories (Ethics, Risk, Sustainability, Equality/Inclusion, Security). Costs missing or irrelevant to the project. Lacks understanding of the project realities.
Professional Communication (10%)	Professional standard of writing and presentation. Clear, concise, and well-structured throughout. Figures and references are consistently well formatted.	Well-written and clearly presented. The structure is logical and easy to follow. Minor lapses in style or formatting.	Readable and generally clear. Structure adequate, though with occasional inconsistency or weak formatting. Some errors in style or referencing.	Basic presentation with uneven writing style. Structure is not always logical; several formatting or referencing errors.	Poorly presented, with unclear writing and weak structure. Figures/tables limited or poorly formatted. Frequent errors.	Very limited attention to presentation or style. Disorganised and difficult to follow. Numerous major errors.

	%	Fail	3: Threshold standard: major and persistent shortcomings	2ii: Satisfactory standard: Several substantial shortcomings	2i: Good standard: some relatively small shortcomings	1st: Excellent standard: Few shortcomings
Aims and Objectives	5	Aims are absent or entirely disconnected from the solution.	Aims are stated but generic, showing a superficial link to the solution.	Aims are relevant but lack technical specificity or measurable outcomes.	Clear, technical objectives that align well with the proposed solution requirements.	Precise, highly focused objectives that demonstrate a sophisticated understanding of the technical scope.
Introduction and Background	10	The report lacks a coherent rationale. There is no evidence of a structure for the work undertaken or a record of executed tasks.	Provides a basic summary of the problem, but the justification for the chosen solution is weak. Includes a skeletal record of work performed that fails to detail how specific milestones were achieved.	A descriptive justification for the solution is provided. Includes an account of the work carried out that identifies the primary stages of development, though the practical transition between tasks is often vague.	A robust, evidence-based rationale that justifies the technical approach. Provides a detailed mapping of the actual work performed, documenting the logical progression from initial requirements to the realised output.	A sophisticated and persuasive technical justification. Includes a comprehensive operational breakdown of the project's lifecycle, clearly identifying completed work streams, individual task ownership, and how the project was adapted to meet technical challenges.
Description of wearable device	10	The solution is not described, or the physical/digital output is missing. There is no viable testing methodology provided.	Provides a basic outline of the solution and testing approach. However, the testing lacks the depth required to verify the device against the original technical specification.	A descriptive summary of the solution is provided, though omissions in technical detail make the final delivery difficult to fully evaluate. A testing method is present but lacks the granularity to confirm if all specifications were met.	A robust description of the solution and testing protocol. Sufficient technical detail is provided to understand the group's delivery. The testing method is structured to address the majority of the technical specifications, though some minor gaps in validation remain.	A comprehensive and highly professional exposition of the realised solution. The testing methodology is systematic, rigorous, and provides clear, documented evidence that the prototype meets (or exceeds) the technical specifications defined at the project's outset.
Analysis of Effectiveness	10	Evaluation is absent or superficial. Test data is not presented or is disconnected from the discussion. No	Provides a basic report on prototype performance. Analysis is largely descriptive and qualitative, with conclusions that lack a clear evidentiary link to	Reports on effectiveness with appropriate analytical methods. Conclusions identify basic strengths and weaknesses but remain qualitative. Includes a summary of prototype and	A critical evaluation of the prototype's effectiveness. Quantitative results are presented clearly and used to support valid conclusions. Includes a detailed and largely complete breakdown of project	A sophisticated critical appraisal of prototype performance. Quantitative data is synthesised to provide robust, evidence-led conclusions regarding all design constraints. Includes a comprehensive and meticulous

		consideration of cost.	test results.	testing costs.	expenditure.	breakdown of all prototype and operational costs.
Future Plans	5	No plan is outlined to develop the prototype into a marketable product.	A plan is outlined to develop the prototype into a marketable product. The plan does not address the weaknesses of the prototype.	A plan is presented to develop the prototype into a marketable product that addresses some of the identified weaknesses in the prototype or addresses additional considerations, such as to where the device will sit in the regulatory framework, are addressed but lack detail.	A clear plan is presented to develop the prototype into a marketable product that includes a significant number of the identified weaknesses in the prototype. Additional considerations, such as to where the device will sit in the regulatory framework, are addressed but lack detail.	A clear plan is presented to develop the prototype into a marketable product that includes almost all of the identified weaknesses in the prototype. Additional considerations, such as to where the device will sit in the regulatory framework, are clearly addressed.
Individual Contribution	35	Individual contribution is either absent or lacks sufficient evidence to verify the author's role in the technical development of the project.	Provides an outline of personal involvement, but task ownership remains ambiguous within the group context. Evidence of the contribution's evolution is minimal; recovery plans for incomplete tasks lack technical substance.	Clearly identifies the author's assigned tasks. Contribution is supported by primary evidence (e.g., logbooks), though the description of the work's progression is largely narrative. Plans for incomplete work cover the main tasks but lack fine technical detail.	A robust account of significant individual technical impact. The evolution of the work is clearly documented with a strong audit trail. Where progress was hindered, a technically sound and detailed plan for completion is presented, covering all significant outstanding tasks.	Exemplary demonstration of individual technical agency and mastery. The contribution is fully validated through a meticulous audit trail (e.g., logbooks, version control, or schematics). Provides a sophisticated, comprehensive, and technically precise strategy for the resolution of all outstanding work.
Conclusions, Reflective Summary, and Strengths Analysis	15	Conclusions are absent or fundamentally disconnected from the report's evidence. No meaningful reflection on project management or personal development is	A superficial conclusion that overlooks significant project outcomes. Reflection on project skills is descriptive rather than analytical, with a lack of evidence (e.g. missing Gantt charts or York Strengths data).	Provides a summary conclusion of the project results. The reflective discussion identifies responses to unforeseen problems but remains generic. Reflection is supported by limited evidence.	A robust conclusion that synthesises the report's key findings. The reflection is critically evaluative, identifying specific lessons learned in project planning and management. Multiple pieces of evidence (Gantt, York Strengths, etc.) are used to support the analysis.	An insightful conclusion that provides a comprehensive synthesis of the project. The reflection is reflexive and sophisticated, offering a deep analysis of personal and group professional growth. All required evidence (Gantt, WBS, York Strengths) is seamlessly integrated to justify a rigorous plan for future professional improvement.

		provided.				
Overall Appearance and Structure	10	The report lacks a coherent structure. Pervasive grammatical and spelling errors significantly impede meaning. Visual aids are absent or entirely disconnected from the text.	Structure is fragmented, making the logical flow difficult to follow. Referencing is attempted but inconsistent. Figures are present but lack clarity or integration, and errors in syntax frequently obscure the technical content.	Logical sectioning is evident, though further sub-structuring would improve clarity. Generally free from major errors, though minor lapses in technical nomenclature persist. Figures are appropriate, but appendices are poorly organised or underutilised.	A professionally structured report with effective use of subsections. All sources are appropriately cited using standard academic conventions. Figures and diagrams are of high quality, integrated with clear captions, and cross-referenced within the text. Appendices are relevant and well-curated.	Exemplary structural logic and professional presentation. The prose is precise, using sophisticated technical nomenclature with almost no grammatical errors. Visual data and appendices are seamlessly integrated to support the narrative, demonstrating a high level of communication mastery.